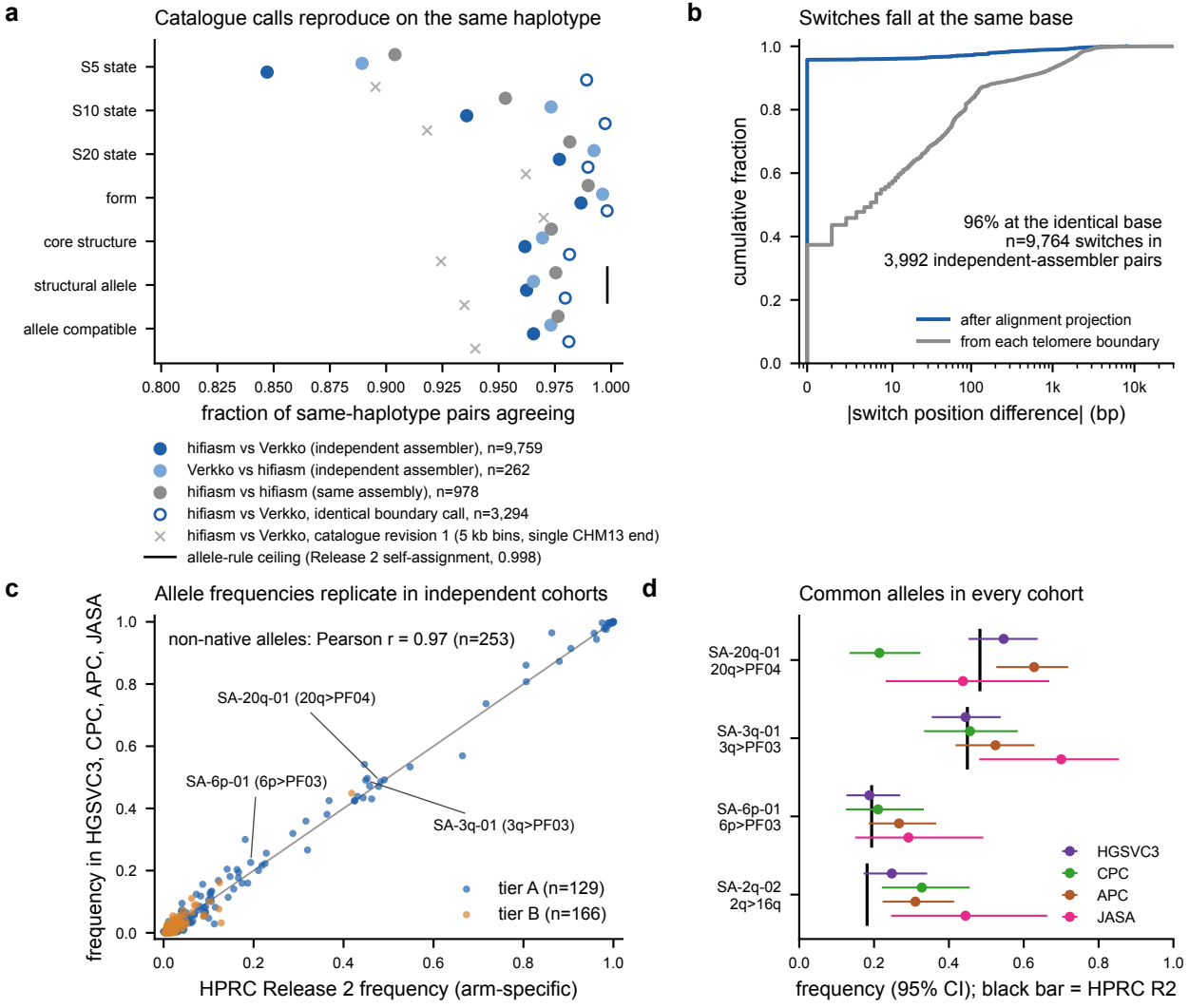

a
Catalogue calls reproduce on the same haplotype
S5 state
S10 state
S20 state
form
core structure
structural allele
allele compatible
0.800 0.825 0.850 0.875 0.900 0.925 0.950 0.975 1.000
fraction of same-haplotype pairs agreeing
hifiasm vs Verkko (independent assembler), n=9,759
Verkko vs hifiasm (independent assembler), n=262
hifiasm vs hifiasm (same assembly), n=978
hifiasm vs Verkko, identical boundary call, n=3,294
hifiasm vs Verkko, catalogue revision 1 (5 kb bins, single CHM13 end)
allele-rule ceiling (Release 2 self-assignment, 0.998)
b
Switches fall at the same base
cumulative fraction
0.0 0.2 0.4 0.6 0.8 1.0
96% at the identical base
n=9,764 switches in
3,992 independent-assembler pairs
after alignment projection
from each telomere boundary
0 10 100 1k 10k
|switch position difference| (bp)
c
Allele frequencies replicate in independent cohorts
non-native alleles: Pearson r = 0.97 (n=253)
SA-20q-01 (20q>PF04)
SA-6p-01 (6p>PF03)
SA-3q-01 (3q>PF03)
tier A (n=129)
tier B (n=166)
frequency in HGSVC3, CPC, APC, JASA
0.0 0.2 0.4 0.6 0.8 1.0
HPRC Release 2 frequency (arm-specific)
d
Common alleles in every cohort
SA-20q-01
20q>PF04
SA-3q-01
3q>PF03
SA-6p-01
6p>PF03
SA-2q-02
2q>16q
HGSVC3
CPC
APC
JASA
0.0 0.2 0.4 0.6 0.8 1.0
frequency (95% CI); black bar = HPRC R2